

CLOUD COMPUTING & BIG DATA ANALYTICS

CONCEPT MAPS:

Q.1) Cloud computing Fundamentals. Include definition, characteristics, benefits and limitations.

Definition:

Cloud computing is the on-demand delivery of computing services such as servers, storage, databases, networking, software and processing power over the internet ("the cloud"), allowing users to access resources on demand without owning or managing physical infrastructure.

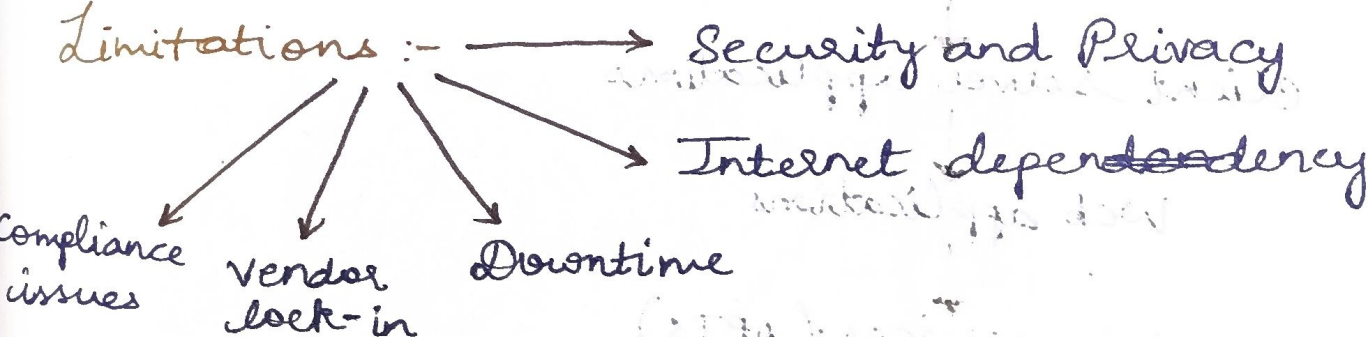
Benefits:-

- Internet Based
- Cost Savings
- Scalability
- Flexibility
- High availability
- Easy maintenance
- Disaster recovery
- Collaboration
- Pay-as-you-use

Characteristics:-

- On demand self service
- Broad access network
- Resource pooling
- Rapid elasticity
- Measured service.

Limitations:-



Q.2) Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems and service delivery delivery.

Hardware evolution:

Mainframe Computers

Large, centralized systems serving many users



Personal Computers (PCs)

Individual computing became common



Servers

Dedicated machines provided network services



Virtualization

Multiple virtual machines run on 1 physical server



Data centers

Large collections of servers store and process data



Cloud Infrastructure

Global data centers deliver computing resources over the internet.

Software evolution:-

Standalone software



Client server applications



Web applications



Web services (APIs)



Cloud Applications (SaaS)

Distributed Computing

Cluster computing

Grid computing

Cloud computing

Service delivery :-

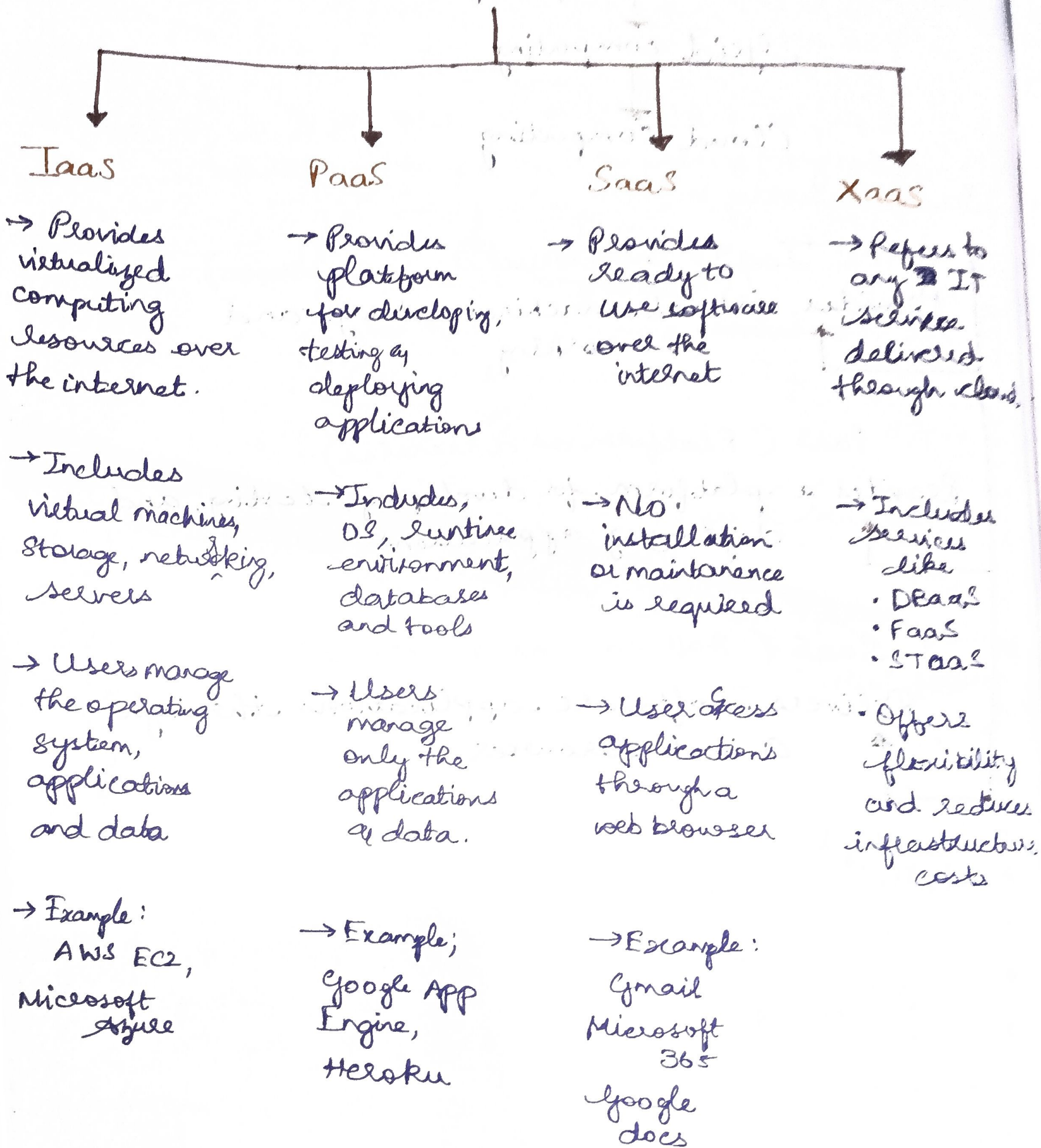
IaaS (Infrastructure as a service)
Provides virtual machines, storage and networking

PaaS (Platform as a service)
Provides a platform for developing, testing, and deploying applications.

SaaS (Software as a service)
Delivers software applications through a web browser.

Q.3). Service models. Map IaaS, PaaS, SaaS and XaaS with provider responsibilities and user responsibilities.

CLOUD SERVICE MODELS



Q.4) Virtualization and Web services as enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

